

Results

Note that this is an ongoing list.

Oct 3, 2026 · @Shrey

The CFON* chromatic number of $K(n, \kappa)$, written χ below, is known exactly for every $n \geq 5$ when $\kappa = 2$ and for $n = 7$ and 8 when $\kappa = 3$. The conjectured value $\kappa + 1$ fails at seven pairs (n, κ) .

Values

Graph	Vertices	Conjectured χ	Established χ	Evidence
$K(5,2)$	10	3	2	Explicit 2-coloring
$K(6,2)$	15	3	2	Explicit 2-coloring
$K(7,2)$	21	3	3	2 colors fail, with verified DRAT proof
$K(8,2)$	28	3	3	2 colors fail, with verified DRAT proof
$K(9,2)$	36	3	3	2 colors fail, with verified DRAT proof
$K(n,2), n \geq 10$	$n(n-1)/2$	3	3	Proved in the reference paper
$K(7,3)$	35	4	2	Explicit 2-coloring
$K(8,3)$	56	4	2	Explicit 2-coloring
$K(9,3)$	84	4	2 or 3	Explicit 3-coloring; 2 colors unresolved
$K(10,3)$	120	4	2 or 3	Explicit 3-coloring; 2 colors unresolved
$K(11,3)$	165	4	2 or 3	Explicit 3-coloring; 2 colors unresolved

Graph	Vertices	Conjectured χ	Established χ	Evidence
$K(n,3), 12 \leq n \leq 20$	220 to 1140	4	Open, between 2 and 4	Not yet computed
$K(n,3), n \geq 21$	$n(n-1)(n-2)/6$	4	4	Proved in the reference paper

The lower bound of 2 in every row comes from the theorem that one color never suffices when $n \geq 2k + 1$. The upper bound of $k + 1$ comes from the construction in the reference paper. The theorem and its proof are in [paper /](#) .

Method

Each question "can $K(n, k)$ be CFON*-colored with k colors?" is translated into a Boolean satisfiability (SAT) instance and given to a solver. The instance is satisfiable exactly when such a coloring exists.

Encoding

For each vertex v and each color c from 1 to k there are two variables. $x(v, c)$ means v has color c . $u(v, c)$ means c is the color that v sees exactly once among its neighbors. A vertex with no x variable true is uncolored.

1. Each vertex has at most one color: for every pair of colors c and c' , $x(v, c)$ and $x(v, c')$ are not both true.
2. Each vertex sees some color exactly once: at least one of $u(v, 1), \dots, u(v, k)$ is true.
3. If $u(v, c)$ is true, then exactly one neighbor w of v has $x(w, c)$ true. The exactly-one condition uses a sequential-counter cardinality encoding.

Solving and checking

Instances are built with PySAT and solved with CaDiCaL. Every answer, SAT or UNSAT is verified via a second layer.

Solver answer	Meaning	Independent check
Satisfiable	k colors suffice; the solver returns a coloring	A separate script rebuilds the graph from the definition and tests the CFON* condition at every vertex. It shares no code with the encoding.
Unsatisfiable	k colors do not suffice	The solver (Glucose 3) emits a DRAT proof, which drat-trim verifies against the CNF formula.

The encoding itself was validated against brute-force enumeration on $K(5,2)$ and $K(6,2)$, where every assignment can be tested directly.

The CNF formulas and DRAT proofs are in `certificates/`. The scripts are in `code/`.